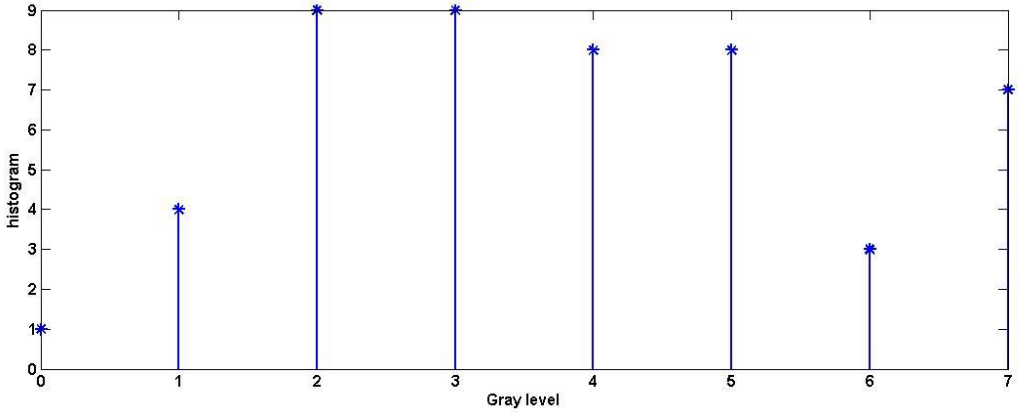
No.1

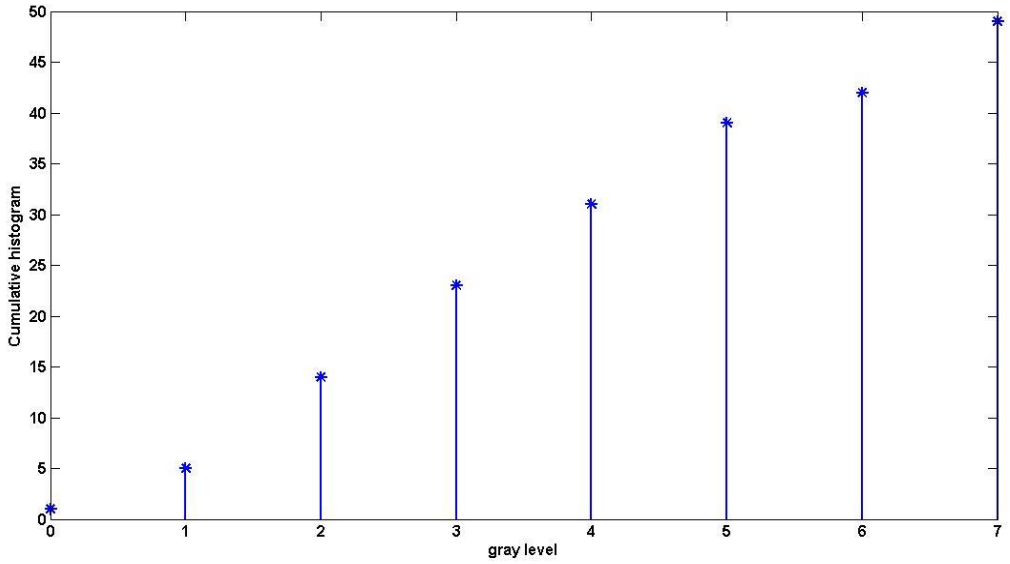
Gray level G1(x,y)	Histogram N (g1(x,y))	Cumulative histogram C g1(x,y)	Temp= (K*Cg1(x,y)/MN)-1	G2(x,y)= ceiling (temp)
0	1	1	-0.836	0
1	4	5	-0.183	0
2	9	14	1.28	2
3	9	23	2.75	3
4	8	31	4.06	5
5	8	39	5.36	6
6	3	42	5.85	6
7	7	49	6.67	7

Where k=8 gray level , MN=7*7=49

Histogram



Cumulative histogram



Replace old gray level with new gray level

So new image

2	7	6	5	7	3	2
7	2	6	5	3	6	0
5	3	2	6	0	5	3
5	3	7	0	2	6	7
3	2	6	5	3	2	6
6	7	5	3	2	6	5
6	3	0	7	6	0	2

10% of dark rerigon :

I have 49 pixel so no of pixels=5 so eliminate graylevel 0 and 1

20% of lihght region

I have 49 pixel so no of pixels=10 so eliminate graylevel 6 and 7

The new image contain gray level from 2 till 5

Min and max of image =2,5 max and min of required stretching 0 and 7

So we have 2 points (2,0) (5,7) so apply line equation

(Y2-y1)/(x2-x1)=(y-y1)/(x-x1)

Y=new gray level

X=old gray level

New gray level=7 (old gray level -2)/3=7/3*old gray level-14/3

Y=ax+b (a=7/3,b= −14/3

New image

0	7	7	5	7	3	0
7	0	7	5	3	7	0
5	3	0	7	0	5	3
5	3	7	0	0	7	7
3	0	7	5	3	0	7
7	7	5	3	0	7	5
7	3	0	7	7	0	0

No. 2

3	4	6
4	5	6
2	3	5

There are two ways to solve this problem

Soultion-1

$$g_2(x,y)=\frac{d-c}{b-a}(g_1(x,y)-a)+c$$

Where $g_2(x,y)$: modified image (required image image)

$g_1(x,y)$:Old image (given image)

(a,b): range of the old image (c,d) range of required image

In this problem

Step-1: a=2,b=6,c=0,d=7

Step-2:

$$g_2(x,y)=\frac{7-0}{6-2}(g_1(x,y)-2)+0$$

$$g_2(x,y)=\frac{7}{4}(g_1(x,y)-2)$$

Step-3:

Aply on each pixel to formulate new image

At pixel=3

g2 =7/4(3-2)=7/4=1.75=2

and so on for all pixels (note must round results)

step-4:

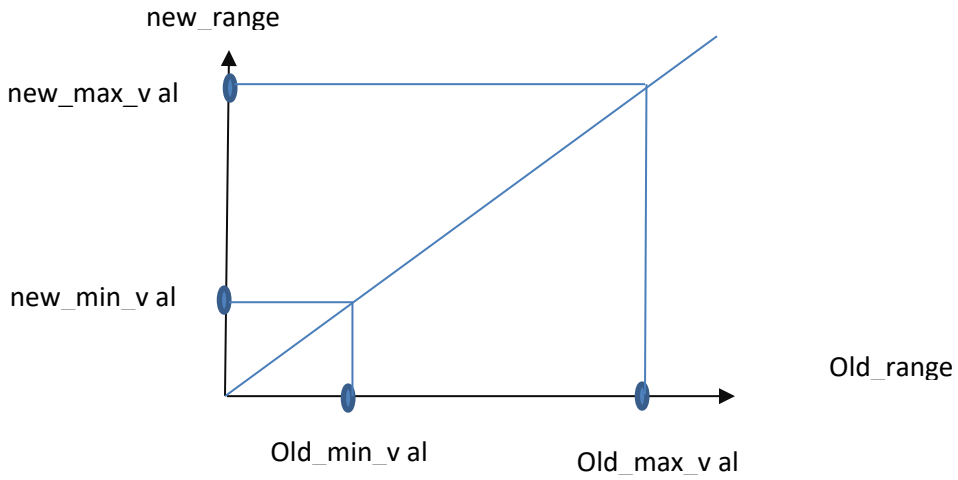
Formulate new image wih new pixel value calculated in step 3

~~3~~ (2)	4	6
4	5	6
2	3	5

Soultion-2

We only consider linear contrast stretching so we want to transform the given image with (old_min_val and old_max_val) into a new range (new_min_val and new_max_val) using linear equation. To formulate line equation we need two points

So



For this problem

(old_min_val,new_min_val)=(x1,y1)=(2,0)

(old_max_val,new_max_val)=(x2,y2)=(6,7)

So line equation will be :

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

$$\frac{7-0}{6-2} = \frac{y-0}{x-2}$$

$$4(y) = 7(x-2)$$

$$y = 7/4(x-2)$$

Continue as pervious solution (step 3 and 4)